2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

3.0 TDV6 ENGINE CONTROL MODULE (ECM) - B10A2-01 TO P02D7-32

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places and with a current calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed via the 'diagnostic trouble code monitor' tab on the manufacturers approved diagnostic system
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

The table below lists all diagnostic trouble codes (DTCs) that could be logged in the rear integrated control panel, for additional diagnosis and testing information refer to the relevant diagnosis and testing section. For additional information, refer to: Electronic Engine Controls (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B10A2-01	Crash Input - General electrical failure	**NOTE:** Purpose of the DTC. To indicate that the hardware signal received from restraints control module is unreliable ▪ This DTC may be set because of vehicle collision ▪ The hardwired crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Other restraints control module related DTCs ▪ CAN harness link between engine control module and restraints control module network malfunction ▪ Restraints control module power and ground circuits open circuit	▪ Refer to the electrical circuit diagra and check the hardwired crash sign circuit for short circuit to ground, sl circuit to power, open circuit, high resistance ▪ Check restraints control module fo related DTCs and refer to relevant index ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test ▪ Refer to the electrical circuit diagra and check restraints control module power and ground circuits for oper circuit ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
B10A2-07	Crash Input - Mechanical failures		▪ Refer to the electrical circuit diagra and check the hardwired crash sign circuit for short circuit to ground, sl

			NOTE: Purpose of the DTC. To indicate that the hardware signal received from restraints control module is unreliable - This DTC may be set because of vehicle collision - The hardwired crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Other restraints control module related DTCs - CAN harness link between engine control module and restraints control module network malfunction - Restraints control module power and ground circuits open circuit	circuit to power, open circuit, high resistance - Check restraints control module for related DTCs and refer to relevant index - Using the manufacturer approved diagnostic system, perform a CAN network integrity test - Refer to the electrical circuit diagra and check restraints control module power and ground circuits for open circuit - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	B10A2-26	Crash Input - Signal rate of change below threshold	**NOTE:** Purpose of the DTC. To indicate a duty cycle out of range - The signal transitions more slowly than is reasonably allowed	- Refer to the electrical circuit diagra and check the hardwired crash sign circuit for short circuit to ground, sl circuit to power, open circuit, high resistance - Check restraints control module for related DTCs and refer to relevant index - Using the manufacturer approved diagnostic system, perform a CAN network integrity test - Refer to the electrical circuit diagra

			- This DTC may be set because of vehicle collision - The hardwired crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Other restraints control module related DTCs - CAN harness link between engine control module and restraints control module network malfunction - Restraints control module power and ground circuits open circuit	and check restraints control module power and ground circuits for open circuit - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	B10A2-27	Crash Input - Signal rate of change above threshold	**NOTE:** Purpose of the DTC. To indicate a duty cycle out of range - The signal transitions more quickly than is reasonably allowed - This DTC may be set because of vehicle collision - The hardwired crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Other restraints control module related DTCs	- Refer to the electrical circuit diagra and check the hardwired crash sign circuit for short circuit to ground, sh circuit to power, open circuit, high resistance - Check restraints control module for related DTCs and refer to relevant index - Using the manufacturer approved diagnostic system, perform a CAN network integrity test - Refer to the electrical circuit diagra and check restraints control module power and ground circuits for open circuit - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			• CAN harness link between engine control module and restraints control module network malfunction • Restraints control module power and ground circuits open circuit		
	B10A2-32	Crash Input - Signal low time < minimum	• The engine control module detected the low pulse is too narrow with respect to time • This DTC may be set because of vehicle collision • The hardwired crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance • Other restraints control module related DTCs • CAN harness link between engine control module and restraints control module network malfunction • Restraints control module power and ground circuits open circuit	• Refer to the electrical circuit diagra and check the hardwired crash sign circuit for short circuit to ground, sh circuit to power, open circuit, high resistance • Check restraints control module for related DTCs and refer to relevant index • Using the manufacturer approved diagnostic system, perform a CAN network integrity test • Refer to the electrical circuit diagra and check restraints control module power and ground circuits for open circuit • Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	B10A2-35	Crash Input - Signal high time > maximum	• This DTC is set when the engine control module detects that the signal high time is greater than the maximum value • This DTC may be set because of vehicle collision	• Refer to the electrical circuit diagra and check the hardwired crash sign circuit for short circuit to ground, sh circuit to power, open circuit, high resistance • Check restraints control module for related DTCs and refer to relevant index

			▪ The hardwired crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Other restraints control module related DTCs ▪ CAN harness link between engine control module and restraints control module network malfunction ▪ Restraints control module power and ground circuits open circuit	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test ▪ Refer to the electrical circuit diagra and check restraints control module power and ground circuits for oper circuit ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	B1206-68	Crash Occurred - Event information	**NOTE:** Purpose of the DTC. To indicate a crash has been detected by the engine control module ▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ This DTC may be set because of vehicle collision ▪ Engine control module received	▪ Check restraints control module foi related DTCs and refer to relevant index ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test ▪ Refer to the electrical circuit diagra and check restraints control module power and ground circuits for oper circuit ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			crash event via CAN bus or hardwired input signal 500HZ follow by 250 HZ pattern - Loss of communications with the central junction box via network communications and loss of the PWM signal from the restraints control module - Other restraints control module related DTCs - CAN harness link between engine control module and restraints control module network malfunction - Restraints control module power and ground circuits open circuit	
	C0031-29	Left Front Wheel Speed Sensor - Signal invalid	- The value of the signal measured by the engine control module is not plausible given the operating conditions - Anti-lock brake system failure	- Check anti-lock brake system contr module for related DTCs and refer relevant DTC index - Refer to the electrical circuit diagra and check the power and ground connections to the anti-lock brake system control module - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	C0034-29	Right Front Wheel Speed Sensor - Signal invalid	- The value of the signal measured by the engine control module is not plausible given the	- Check anti-lock brake system contr module for related DTCs and refer relevant DTC index - Refer to the electrical circuit diagra and check the power and ground

			operating conditions ▪ Anti-lock brake system failure	connections to the anti-lock brake system control module ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	C0037-29	Left Rear Wheel Speed Sensor - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Anti-lock brake system failure	▪ Check anti-lock brake system contr module for related DTCs and refer relevant DTC index ▪ Refer to the electrical circuit diagra and check the power and ground connections to the anti-lock brake system control module ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	C003A-29	Right Rear Wheel Speed Sensor - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Anti-lock brake system failure	▪ Check anti-lock brake system contr module for related DTCs and refer relevant DTC index ▪ Refer to the electrical circuit diagra and check the power and ground connections to the anti-lock brake system control module ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0016-3A	Crankshaft Position - Camshaft Position Correlation Bank 1 Sensor A - Incorrect has too many pulses	▪ Other related DTCs ▪ Engine assembled incorrectly ▪ Loose camshaft position sensor ▪ Loose crankshaft position sensor ▪ Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Crankshaft position	▪ Check engine control module for related DTCs and refer to relevant index ▪ Check camshaft position sensor is installed correctly ▪ Check crankshaft position sensor fo correct installation ▪ Check crankshaft position sensor ai gap to target rotor is correct, remo any debris from sensor face, check damaged teeth on rotor ▪ Refer to the electrical circuit diagra and check camshaft position senso circuit for short circuit to ground, sh circuit to power, open circuit, high

			sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Crankshaft position sensor incorrectly installed ▪ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ▪ Camshaft or crankshaft position sensor reluctor ring air gap excessive ▪ Timing chain stretch beyond a tolerable limit ▪ Valve timing incorrect	resistance. Check crankshaft positic sensor circuit for short circuit to grc short circuit to power, open circuit, resistance ▪ Check reluctor ring to sensor runou and air gap are within specification ▪ Refer to the relevant section of the workshop manual and check valve timing is within specification ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0030-29	HO2S Heater Control Circuit Bank 1 Sensor 1 - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions
			▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Heated oxygen sensor failure
				▪ Refer to the electrical circuit diagra and check the heated oxygen sens heater circuit for short circuit to grc short circuit to power, open circuit, resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new heated oxy sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Oxygen sensor' routine (0x403
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatior routine (0x402C)
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0031-11	HO2S Heater Control Circuit Low Bank 1 Sensor 1 - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected
			▪ Heated oxygen sensor heater circuit short circuit to ground
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Heated oxygen sensor failure
				▪ Refer to the electrical circuit diagra and check the heated oxygen sens heater circuit for short circuit to gr
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new heated oxy sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Oxygen sensor' routine (0x403
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatio routine (0x402C)
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0032-12	HO2S Heater Control Circuit High Bank 1 Sensor 1 - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected
			▪ Heated oxygen sensor heater circuit short circuit to power
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Heated oxygen sensor failure
				▪ Refer to the electrical circuit diagra and check the heated oxygen sens heater circuit for short circuit to po
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new heated oxy sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Oxygen sensor' routine (0x403
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatio routine (0x402C)
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0045-13	Turbocharger/Supercharger Boost Control "A"	▪ The engine control module has	▪ Diagnosis of this DTC may require the manufacturer approved diagno

		Circuit/Open - Circuit open	determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output
			▪ Variable geometry turbine vane actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Variable geometry turbine vane actuator failure
				system check datalogger signals
				  ▪ 0x03DF Boost Pressure Actuat Bank 1 - Desired position - %
				  ▪ 0x0532 Commanded Boost Actuator Control Bank 1 - %
				▪ Refer to the electrical circuit diagra and check the variable geometry turbine vane actuator circuit for sho circuit to ground, short circuit to po open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Using the manufacturer approved diagnostic system perform routine Inline diagnostic unit 2 diagnostic t Turbocharger
				▪ Check and install a new variable geometry turbine vane actuator as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Ai path set-up routine' routine's (0x4030/0x4051_01/0x4050_01 0xF405)
				  ▪ Using the manufacturer approv diagnostic system carry out 'En control module low pressure E(adaption clear' routine (0x0406
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0046-16	Turbocharger/Supercharger Boost Control "A" Circuit Range/Performance - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground
			▪ Variable geometry turbine vane actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x03DF Boost Pressure Actuat Bank 1 - Desired position - %
				  ▪ 0x0532 Commanded Boost Actuator Control Bank 1 - %
				▪ Refer to the electrical circuit diagra and check the variable geometry turbine vane actuator circuit for sho circuit to ground, short circuit to po open circuit, high resistance
				▪ Inspect connectors for signs of wat

			- Connector is disconnected, connector pin is backed out, connector pin corrosion - Variable geometry turbine vane actuator failure	ingress, and pins for damage and/c corrosion - Check and install a new variable geometry turbine vane actuator as required - Using the manufacturer approv diagnostic system carry out 'Ai path set-up routine' routine's (0x4030/0x4051_01/0x4050_01 0xF405) - Using the manufacturer approv diagnostic system carry out 'En control module low pressure E(adaption clear' routine (0x040é - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0046-19	Turbocharger/Supercharger Boost Control "A" Circuit Range/Performance - Circuit current above threshold	- The engine control module has measured current flow above a specified range - Variable geometry turbine vane actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Variable geometry turbine vane actuator failure	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x03DF Boost Pressure Actuato Bank 1 - Desired position - % 0x0532 Commanded Boost Actuator Control Bank 1 - % - Refer to the electrical circuit diagra and check the variable geometry turbine vane actuator circuit for sho circuit to ground, short circuit to po open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new variable geometry turbine vane actuator as required - Using the manufacturer approv diagnostic system carry out 'Ai path set-up routine' routine's (0x4030/0x4051_01/0x4050_01 0xF405) - Using the manufacturer approv diagnostic system carry out 'En control module low pressure E(adaption clear' routine (0x040é - Using the manufacturer approved diagnostic system clear all stored D

				using the Diagnosis Menu tab and retest

	P0046-1D	Turbocharger/Supercharger Boost Control "A" Circuit Range/Performance - Circuit current out of range	▪ The engine control module has detected a current outside of the expected range, but not identified as too high or too low
			▪ Variable geometry turbine vane actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Variable geometry turbine vane actuator failure
				▪ Diagnosis of this DTC may require
				the manufacturer approved diagno
				system check datalogger signals
				  ▪ 0x03DF Boost Pressure Actuat
				  Bank 1 - Desired position - %
				  ▪ 0x0532 Commanded Boost Actuator Control Bank 1 - %
				▪ Refer to the electrical circuit diagra
				and check the variable geometry
				turbine vane actuator circuit for sh
				circuit to ground, short circuit to p
				open circuit, high resistance
				▪ Inspect connectors for signs of wat
				ingress, and pins for damage and/
				corrosion
				▪ Check and install a new variable geometry turbine vane actuator as required
				  ▪ Using the manufacturer approv
				  diagnostic system carry out 'Ai
				  path set-up routine' routine's
				  (0x4030/0x4051_01/0x4050_01
				  0xF405)
				  ▪ Using the manufacturer approv
				  diagnostic system carry out 'En
				  control module low pressure E
				  adaption clear' routine (0x040
				▪ Using the manufacturer approved
				diagnostic system clear all stored D
				using the Diagnosis Menu tab and
				retest

	P0046-74	Turbocharger/Supercharger Boost Control "A" Circuit Range/Performance - Actuator slipping	▪ Turbocharger vane actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Refer to the electrical circuit diagra and check the turbocharger vane actuator circuit for short circuit to ground, short circuit to power, ope circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/ corrosion ▪ Using the manufacturer approved diagnostic system perform routine Inline diagnostic unit 2 diagnostic t Turbocharger

			▪ Turbocharger vane actuator failure	▪ Check and install a new turbocharg vane actuator as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0046-77	Turbocharger/Supercharger Boost Control "A" Circuit Range/Performance - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment
			▪ Variable geometry turbine vane actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Variable geometry turbine vane actuator failure
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x03DF Boost Pressure Actuat Bank 1 - Desired position - %
				  ▪ 0x0532 Commanded Boost Actuator Control Bank 1 - %
				▪ Refer to the electrical circuit diagra and check the variable geometry turbine vane actuator circuit for sho circuit to ground, short circuit to po open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Using the manufacturer approved diagnostic system perform routine Inline diagnostic unit 2 diagnostic t Turbocharger
				▪ Check and install a new variable geometry turbine vane actuator as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Ai path set-up routine' routine's (0x4030/0x4051_01/0x4050_01 0xF405)
				  ▪ Using the manufacturer approv diagnostic system carry out 'En control module low pressure E(adaption clear' routine (0x0406
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0047-11	Turbocharger/Supercharger Boost Control "A" Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a	▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x03DF Boost Pressure Actuato

			period longer than expected or has detected a ground measurement when another value was expected ▪ Variable geometry turbine vane actuator circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Variable geometry turbine vane actuator failure	Bank 1 - Desired position - % ▪ 0x0532 Commanded Boost Actuator Control Bank 1 - % ▪ Refer to the electrical circuit diagra and check the variable geometry turbine vane actuator circuit for sho circuit to ground ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Using the manufacturer approved diagnostic system perform routine Inline diagnostic unit 2 diagnostic t Turbocharger ▪ Check and install a new variable geometry turbine vane actuator as required ▪ Using the manufacturer approv diagnostic system carry out 'Ai path set-up routine' routine's (0x4030/0x4051_01/0x4050_01 0xF405) ▪ Using the manufacturer approv diagnostic system carry out 'En control module low pressure E(adaption clear' routine (0x040é ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0048-12	Turbocharger/Supercharger Boost Control "A" Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Variable geometry turbine vane actuator circuit short circuit to power ▪ Connector is disconnected, connector pin is	▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x03DF Boost Pressure Actuato Bank 1 - Desired position - % ▪ 0x0532 Commanded Boost Actuator Control Bank 1 - % ▪ Refer to the electrical circuit diagra and check the variable geometry turbine vane actuator circuit for sho circuit to power ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Using the manufacturer approved diagnostic system perform routine

			backed out, connector pin corrosion
			▪ Variable geometry turbine vane actuator failure
				Inline diagnostic unit 2 diagnostic t Turbocharger
				▪ Check and install a new variable geometry turbine vane actuator as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Ai path set-up routine' routine's (0x4030/0x4051_01/0x4050_01 0xF405)
				  ▪ Using the manufacturer approv diagnostic system carry out 'En control module low pressure E(adaption clear' routine (0x0406
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P004A-13	Turbocharger/Supercharger Boost Control "B" Circuit/Open - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output
			▪ Secondary turbocharger charge air shut-off solenoid circuit open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Secondary turbocharger charge air shut-off solenoid failure
				▪ Refer to the electrical circuit diagra and check the secondary turbochar charge air shut-off solenoid circuit f open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new secondary turbocharger charge air shut-off solenoid as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Ai path set-up routine' routine's (0x4030/0x4051_01/0x4050_01 0xF405)
				  ▪ Using the manufacturer approv diagnostic system carry out 'En control module low pressure E(adaption clear' routine (0x0406
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P004B-77	Turbocharger/Supercharger Boost Control "B" Circuit Range/Performance - Commanded position not	▪ Secondary turbocharger charge air shut-off valve system mechanical	▪ Check the vacuum system, Charge shut-off valve and Turbine intake sh off valve operation. **Refer to sectic 303-04B or 303-04D and perform**

		reachable	integrity ▪ Secondary turbocharger charge air shut-off valve system vacuum leakage	**pinpoint test A Vacuum Control System Tests** ▪ Using the Jaguar Land Rover Appr Diagnostic Equipment, clear all sto DTCs and retest

	P004C-11	Turbocharger/Supercharger Boost Control "B" Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected
			▪ Secondary turbocharger charge air shut-off solenoid circuit short circuit to ground, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Secondary turbocharger charge air shut-off solenoid failure
				▪ Refer to the electrical circuit diagra and check the secondary turbochar charge air shut-off solenoid circuit f short circuit to ground, open circuit high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new secondary turbocharger charge air shut-off solenoid as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Ai path set-up routine' routine's (0x4030/0x4051_01/0x4050_01 0xF405)
				  ▪ Using the manufacturer approv diagnostic system carry out 'En control module low pressure EC adaption clear' routine (0x0406
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P004C-16	Turbocharger/Supercharger Boost Control "B" Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Secondary turbocharger charge air shut-off solenoid circuit short circuit to ground, open circuit, high resistance ▪ Connector is	▪ Refer to the electrical circuit diagra and check the secondary turbochar charge air shut-off solenoid circuit f short circuit to ground, open circuit high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new secondary turbocharger charge air shut-off solenoid as required ▪ Using the manufacturer approv diagnostic system carry out 'Ai path set-up routine' routine's

			disconnected, connector pin is backed out, connector pin corrosion ▪ Secondary turbocharger charge air shut-off solenoid failure	(0x4030/0x4051_01/0x4050_01 0xF405) ▪ Using the manufacturer approv diagnostic system carry out 'En control module low pressure EC adaption clear' routine (0x0406 ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P004D-12	Turbocharger/Supercharger Boost Control "B" Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Secondary turbocharger charge air shut-off solenoid circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Secondary turbocharger charge air shut-off solenoid failure	▪ Refer to the electrical circuit diagra and check the secondary turbochar charge air shut-off solenoid circuit f short circuit to power ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new secondary turbocharger charge air shut-off solenoid as required ▪ Using the manufacturer approv diagnostic system carry out 'Ai path set-up routine' routine's (0x4030/0x4051_01/0x4050_01 0xF405) ▪ Using the manufacturer approv diagnostic system carry out 'En control module low pressure EC adaption clear' routine (0x0406 ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P004D-17	Turbocharger/Supercharger Boost Control "B"Circuit High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Secondary turbocharger charge air shut-off solenoid circuit short circuit to	▪ Refer to the electrical circuit diagra and check the secondary turbochar charge air shut-off solenoid circuit f short circuit to power ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new secondary turbocharger charge air shut-off solenoid as required ▪ Using the manufacturer approv

			power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Secondary turbocharger charge air shut-off solenoid failure	diagnostic system carry out 'Ai path set-up routine' routine's (0x4030/0x4051_01/0x4050_01 0xF405) - Using the manufacturer approv diagnostic system carry out 'En control module low pressure E adaption clear' routine (0x0406 - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P006A-84	MAP - Mass or Volume Air Flow Correlation Bank 1 - Signal below allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range - Intake air induction system leakage - Manifold absolute pressure and temperature sensor circuit short circuit to ground - Connector is disconnected, connector pin is backed out, connector pin corrosion - Manifold absolute pressure and temperature sensor failure	- Check air induction system for leak - Refer to the electrical circuit diagra and check the manifold absolute pressure and temperature sensor ci for short circuit to ground - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new manifold absolute pressure and temperature sensor as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P006A-85	MAP - Mass or Volume Air Flow Correlation Bank 1 - Signal above allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Charged air	- Check air induction system for leak - Refer to the electrical circuit diagra and check the manifold absolute pressure and temperature sensor ci for short circuit to power - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion

			- induction system leakage - Manifold absolute pressure and temperature sensor circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Manifold absolute pressure and temperature sensor failure	- Check and install a new manifold absolute pressure and temperature sensor as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0071-64	Ambient Air Temperature Sensor Circuit "A" Range/Performance - Signal plausibility failure	- The engine control module detected plausibility failures - Ambient air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Ambient air temperature sensor failure	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x03BA Ambient Air Temperat Sensor Voltage - Volts 0xF446 Ambient Air Temperatu Deg C - Refer to the electrical circuit diagra and check the ambient air tempera sensor circuit for short circuit to gro short circuit to power, open circuit, resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new ambient ai temperature sensor as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0071-84	Ambient Air Temperature Sensor Circuit "A" Range/Performance - Signal below allowable range	The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x03BA Ambient Air Temperat Sensor Voltage - Volts 0xF446 Ambient Air Temperatu Deg C

			▪ range	▪ Refer to the electrical circuit diagra and check the ambient air tempera sensor circuit for short circuit to gr
			▪ Ambient air temperature sensor circuit short circuit to ground	▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Check and install a new ambient ai temperature sensor as required
			▪ Ambient air temperature sensor failure	▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0071-85	Ambient Air Temperature Sensor Circuit "A" Range/Performance - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Ambient air temperature sensor circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Ambient air temperature sensor failure	▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x03BA Ambient Air Temperat Sensor Voltage - Volts ▪ 0xF446 Ambient Air Temperatu Deg C ▪ Refer to the electrical circuit diagra and check the ambient air tempera sensor circuit for short circuit to po ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new ambient ai temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0072-16	Ambient Air Temperature Sensor Circuit "A" Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Ambient air temperature sensor circuit short circuit to ground	▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x03BA Ambient Air Temperat Sensor Voltage - Volts ▪ 0xF446 Ambient Air Temperatu Deg C ▪ Refer to the electrical circuit diagra and check the ambient air tempera sensor circuit for short circuit to gr

			- Connector is disconnected, connector pin is backed out, connector pin corrosion - Ambient air temperature sensor failure	- Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new ambient ai temperature sensor as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0073-17	Ambient Air Temperature Sensor Circuit "A" High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Ambient air temperature sensor circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Ambient air temperature sensor failure	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x03BA Ambient Air Temperat Sensor Voltage - Volts 0xF446 Ambient Air Temperatu Deg C - Refer to the electrical circuit diagra and check the ambient air tempera sensor circuit for short circuit to po - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new ambient ai temperature sensor as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P007A-62	Charge Air Cooler Temperature Sensor Circuit Bank 1 - Signal compare failure	- The engine control module detected failure when comparing two or more input parameters for plausibility - Intake air temperature sensor, charge air cooler temperature sensor, ambient air temperature sensor, manifold absolute pressure and temperature sensor circuit short circuit to	- Allow engine to soak for a period c less than 8 hours, measure and compare sensor temperature value the following sensors - Intake air temperature sensor - Charge air cooler temperature sensor - Ambient air temperature senso - Manifold absolute pressure and temperature sensor - The sensor temperatures values sho be similar to one another - Refer to the electrical circuit diagra and check the intake air temperatu sensor, charge air cooler temperatu

			ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Charge air cooler temperature sensor contamination ▪ Charge air cooler temperature sensor failure	sensor, ambient air temperature se manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check charge air cooler temperatu sensor for contamination ▪ Check and install a new sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P007B-2A	Charge Air Cooler Temperature Sensor Circuit Range/Performance Bank 1 - Signal Stuck In Range	▪ Charge air cooler temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Charge air cooler temperature sensor failure
			  ▪ Contamination
			  ▪ Obstruction
				▪ Refer to the electrical circuit diagra and check the charge air cooler temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new charge air cooler temperature sensor as requi
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P007B-84	Charge Air Cooler Temperature Sensor Circuit Range/Performance Bank 1 - Signal below allowable range	▪ Charge air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagra and check the charge air temperatu sensor circuit for short circuit to gro short circuit to power, open circuit, resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c

			- Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			- Charge air temperature sensor failure
			  - Contamination
			  - Obstruction
				corrosion
				- Check and install a new charge air temperature sensor as required
				- Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P007B-85	Charge Air Cooler Temperature Sensor Circuit Range/Performance Bank 1 - Signal above allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range
			- Charge air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			- Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			- Charge air temperature sensor failure
			  - Contamination
			  - Obstruction
				- Refer to the electrical circuit diagra and check the charge air temperatu sensor circuit for short circuit to gro short circuit to power, open circuit, resistance
				- Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				- Check and install a new charge air temperature sensor as required
				- Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P007C-	Charge Air Cooler	- The engine control	- Refer to the electrical circuit diagra

	16	Temperature Sensor Circuit Low Bank 1 - Circuit voltage below threshold	module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Charge air cooler temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Charge air cooler temperature sensor contamination ▪ Charge air cooler temperature sensor failure	and check the charge air cooler temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check charge air cooler temperatu sensor for contamination ▪ Check and install a new sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P007D-17	Charge Air Cooler Temperature Sensor Circuit High Bank 1 - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Charge air cooler temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Charge air cooler temperature sensor contamination	▪ Refer to the electrical circuit diagra and check the charge air cooler temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check charge air cooler temperatu sensor for contamination ▪ Check and install a new sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

		▪ Charge air cooler temperature sensor failure	
P0087-16	Fuel Rail/System Pressure - Too Low Bank 1 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Other fuel pressure related DTCs ▪ Low pressure fuel system leaks or blockage ▪ Fuel metering valve circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel metering failure	▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index ▪ Check low pressure fuel system for leaks or blockage ▪ Refer to the electrical circuit diagra and check the fuel metering valve circuit for short circuit to ground ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel meteri valve as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
P0087-21	Fuel Rail/System Pressure - Too Low Bank 1 - Signal amplitude < minimum	▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low ▪ Other fuel pressure related DTCs ▪ Low pressure fuel system leaks or blockage ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected,	**NOTE:** Operational requirements needec to allow the monitor to be fully tested. Drive the vehicle over 3 drive cycles (engine on/off) across varied driving conditions includinc high torque to confirm the repair ▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index ▪ Check low pressure fuel system for leaks or blockage ▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground, sl circuit to power, open circuit, high

			connector pin is backed out, connector pin corrosion ▪ Fuel rail pressure sensor failure	resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel rail pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0087-22	Fuel Rail/System Pressure - Too Low Bank 1 - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Other fuel pressure related DTCs ▪ Low pressure fuel system leaks or blockage ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel rail pressure sensor failure	**NOTE:** Operational requirements needec to allow the monitor to be fully tested. Drive the vehicle over 3 drive cycles (engine on/off) across varied driving conditions including high torque to confirm the repair ▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index ▪ Check low pressure fuel system for leaks or blockage ▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel rail pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0087-71	Fuel Rail/System Pressure - Too Low Bank 1 - Signal amplitude < minimum	▪ The engine control module has not detected any motion, in response to energizing a	

			motor, solenoid or relay ▪ Other fuel pressure related DTCs ▪ Low pressure fuel system leaks or blockage ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel rail pressure sensor failure	**NOTE:** Operational requirements neede to allow the monitor to be fully tested. Drive the vehicle over 3 drive cycles (engine on/off) across varied driving conditions includin high torque to confirm the repair ▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index ▪ Check low pressure fuel system for leaks or blockage ▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel rail pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0087-84	Fuel Rail/System Pressure - Too Low Bank 1 - Signal below allowable range	▪ Other fuel pressure related DTCs ▪ Fuel lines restricted ▪ Fuel lines leakage ▪ Fuel injector stuck open / leaking ▪ Fuel pressure control valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out,	▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index ▪ Check fuel lines for restrictions and repair as required ▪ Check fuel lines for leakage and rep as required ▪ Check fuel injector for stuck open / leaking and repair as required ▪ If a new fuel injector is installed Using the manufacturer approv diagnostic system reprogram th injector codes ▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals

			connector pin corrosion ▪ Fuel pressure control valve failure	▪ 0x0324 Fuel Rail Pressure Sens ▪ Refer to the electrical circuit diagra and check the fuel pressure control valve circuit for short circuit to grou short circuit to power, open circuit, resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel pressu control valve as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0088-17	Fuel Rail/System Pressure - Too High Bank 1 - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Other fuel pressure related DTCs ▪ Fuel lines restricted ▪ Fuel lines leakage ▪ Fuel injector stuck open / leaking ▪ Fuel rail pressure sensor circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel rail pressure sensor failure	**NOTE:** Operational requirements needec to allow the monitor to be fully tested. Drive the vehicle over 3 drive cycles (engine on/off) across varied driving conditions including high torque to confirm the repair ▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index ▪ Check fuel lines for restrictions and repair as required ▪ Check fuel lines for leakage and rep as required ▪ Check fuel injector for stuck open / leaking and repair as required ▪ If a new fuel injector is installec Using the manufacturer approv diagnostic system reprogram th injector codes ▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x0324 Fuel Rail Pressure Sens ▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to power

				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel rail pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0088-22	Fuel Rail/System Pressure - Too High Bank 1 - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance ▪ Other fuel pressure related DTCs ▪ Other fuel level sensor related DTCs ▪ Fuel lines restricted ▪ Fuel lines leakage ▪ Fuel rail pressure sensor circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel rail pressure sensor failure	**NOTE:** Operational requirements needec to allow the monitor to be fully tested. Drive the vehicle over 3 drive cycles (engine on/off) across varied driving conditions includin high torque to confirm the repair ▪ Refer to the electrical circuit diagra and check connections are secure a wiring integrity ▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index ▪ Check for fuel level sensor related I and refer to relevant DTC index ▪ Check fuel lines for restrictions and repair as required ▪ Check fuel lines for leakage and rep as required ▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x0324 Fuel Rail Pressure Sens ▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to power ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel rail pressure sensor as required ▪ Using the manufacturer approved

				diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0088-71	Fuel Rail/System Pressure - Too High Bank 1 - Actuator stuck	- The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay - Other fuel pressure related DTCs - Other fuel level sensor related DTCs - Fuel lines restricted - Fuel lines leakage - Low pressure fuel system leaks or blockage - Stuck fuel pump - high pressure - Fuel volume control valve	**NOTE:** Operational requirements needec to allow the monitor to be fully tested. Drive the vehicle over 3 drive cycles (engine on/off) across varied driving conditions including high torque to confirm the repair - Check engine control module for fu pressure related DTCs and refer to relevant DTC index - Check for fuel level sensor related I and refer to relevant DTC index - Check fuel lines for restrictions and repair as required - Check fuel lines for leakage and rep as required - Check low pressure fuel system for leaks or blockage - Check fuel pump - high pressure fo mechanical integrity - Fuel volume control valve - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0088-72	Fuel Rail/System Pressure - Too High Bank 1 - Actuator Stuck Open	- The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to close some piece of equipment - Other fuel pressure related DTCs - Fuel rail pressure sensor circuit short	**NOTE:** Operational requirements needec to allow the monitor to be fully tested. Drive the vehicle over 3 drive cycles (engine on/off) across varied driving conditions including high torque to confirm the repair Check engine control module for fu pressure related DTCs and refer to

			- circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Stuck fuel pump - high pressure - Fuel volume control valve	- relevant DTC index - Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x0324 Fuel Rail Pressure Sens - Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground, sh circuit to power, open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check fuel pump - high pressure fo mechanical integrity - Fuel volume control valve - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0089-4B	Fuel Pressure Regulator 1 Performance - Over temperature	- The engine control module detected an internal temperature above the expected range - Fuel pressure control valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel pressure control valve failure	- Refer to the electrical circuit diagra and check the fuel pressure control valve circuit for short circuit to grou short circuit to power, open circuit, resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new fuel pressu control valve as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0090-13	Fuel Pressure Regulator 1 Control Circuit/Open - Circuit open	The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x03C3 Fuel Pressure Relief Co Valve Duty Cycle - % - Refer to the electrical circuit diagra

			the state of an input in response to an output ▪ Fuel pressure control valve circuit open circuit, high resistance ▪ Fuel pressure control valve failure	and check the fuel pressure control valve circuit for open circuit, high resistance ▪ Check and install a new fuel pressu control valve as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0091-11	Fuel Pressure Regulator 1 Control Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Fuel pressure control valve circuit short circuit to ground ▪ Fuel pressure control valve failure	▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x03C3 Fuel Pressure Relief Co Valve Duty Cycle - % ▪ Refer to the electrical circuit diagra and check the fuel pressure control valve circuit for short circuit to grou ▪ Check and install a new fuel pressu control valve as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0092-12	Fuel Pressure Regulator 1 Control Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Fuel pressure control valve circuit short circuit to power ▪ Fuel pressure control valve failure	▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x03C3 Fuel Pressure Relief Co Valve Duty Cycle - % ▪ Refer to the electrical circuit diagra and check the fuel pressure control valve circuit for short circuit to pow ▪ Check and install a new fuel pressu control valve as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P00BC-00	Mass or Volume Air Flow "A" Circuit Range/Performance - Air Flow Too Low - No sub type information	▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high	▪ Refer to the electrical circuit diagra and check the mass air flow sensor circuit for short circuit to ground, sl circuit to power, open circuit, high resistance

			resistance
			▪ Exhaust gas recirculation valve - High pressure circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Mass air flow sensor failure
			  ▪ Contamination
			  ▪ Obstruction
			▪ Exhaust gas recirculation valve - High pressure failure
			  ▪ Contamination
			  ▪ Obstruction
				▪ Refer to the electrical circuit diagra and check the exhaust gas recircula valve - High pressure circuit for sho circuit to ground, short circuit to po open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new mass air flo sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'En control module mass air flow adaption clear' routine (0x0406
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatior routine (0x402C)
				▪ Check and install a new exhaust ga recirculation valve - High pressure a required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P00BD-00	Mass or Volume Air Flow "A" Circuit Range/Performance - Air Flow Too High - No sub type information	▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas recirculation valve - High pressure circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin	▪ Refer to the electrical circuit diagra and check the mass air flow sensor circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagra and check the exhaust gas recircula valve - High pressure circuit for sho circuit to ground, short circuit to po open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new mass air flo sensor as required ▪ Using the manufacturer approv diagnostic system carry out 'En control module mass air flow adaption clear' routine (0x0406

			corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			- Mass air flow sensor failure
			  - Contamination
			  - Obstruction
			- Exhaust gas recirculation valve - High pressure failure
			  - Contamination
			  - Obstruction
				- Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatio routine (0x402C)
				- Check and install a new exhaust ga recirculation valve - High pressure a required
				- Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P00BE-00	Mass or Volume Air Flow "B" Circuit Range/Performance - Air Flow Too Low - No sub type information	- Intake air system, high pressure boost leak - Air leak at air intake system - Charge air shut-off valve failure - Turbine intake shut-off valve failure	- Check and correct intake air system high pressure boost leak - Check and correct air leak at air int system - Check the vacuum system, Charge shut-off valve and Turbine intake sh off valve operation. **Refer to sectio 303-04B or 303-04D and perform pinpoint test A Vacuum Control System Tests** - Using the Jaguar Land Rover Appro Diagnostic Equipment, clear all sto DTCs and retest
	P00BF-00	Mass or Volume Air Flow "B" Circuit Range/Performance - Air Flow Too High - No sub type information	- Charge air shut-off valve failure - Turbine intake shut-off valve failure - Intake air system, high pressure boost leak	- Check the vacuum system, Charge shut-off valve and Turbine intake sh off valve operation. **Refer to sectio 303-04B or 303-04D and perform pinpoint test A Vacuum Control System Tests** - Check and correct intake air system high pressure boost leak - Using the Jaguar Land Rover Appro Diagnostic Equipment, clear all sto DTCs and retest
	P00C0-13	Turbocharger/Supercharger Bypass Valve "B" Control Circuit - Circuit open	- The engine control module has determined an open circuit via lack of bias voltage, low current	- Refer to the electrical circuit diagra and check the charge air recirculati solenoid circuit for open circuit, hig resistance

			flow, no change in the state of an input in response to an output ▪ Charge air recirculation solenoid circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Charge air recirculation solenoid failure	▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new charge air recirculation solenoid as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P00C1-11	Turbocharger/Supercharger Bypass Valve "B" Control Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Charge air recirculation solenoid circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Charge air recirculation solenoid failure	▪ Refer to the electrical circuit diagra and check the charge air recirculati solenoid circuit for short circuit to ground, open circuit, high resistanc ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new charge air recirculation solenoid as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P00C2-12	Turbocharger/Supercharger Bypass Valve "B" Control Circuit High - Circuit short	▪ The engine control module has detected a vehicle	▪ Refer to the electrical circuit diagra and check the charge air recirculati solenoid circuit for short circuit to

		to battery	power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Charge air recirculation solenoid circuit short circuit to power ▪ Charge air recirculation solenoid failure	power ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P00C6-00	Fuel Rail Pressure Too Low - Engine Cranking Bank 1 - No sub type information	▪ Low level fuel condition ▪ Other fuel pressure related DTCs ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel rail pressure sensor - Check for fuel rail pressure build up during start ▪ Fuel gauge sender unit sticking ▪ Fuel lines restricted ▪ Fuel lines leaking ▪ Fuel pump failure	▪ Check the fuel volume available in t fuel tank is sufficient, add fuel as required. Clear DTC and retest ▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Check fuel system for mechanical integrity ▪ Check the fuel gauge sender units f correct operation ▪ Check fuel lines for restriction ▪ Check fuel lines for leakage ▪ Check for fuel pump related DTCs refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P00CF-62	Barometric Pressure - Turbocharger/Supercharger Boost Sensor "A" Correlation - Signal compare failure	▪ The engine control module detected failure when comparing two or more input parameters for plausibility	▪ This DTC is set when the engine co module detects a signal compare fa on the charge air pressure sensor si Check the sensor hose for blockage leaks. Check the sensor and harnes mechanical damage caused by hea chaffing

			▪ Charge air circuit leak or blockage ▪ Charge air recirculation solenoid failure ▪ Charge air solenoid failure ▪ Mechanical failure - Sensor hose blocked or leaking ▪ Harness failure - Charge air pressure sensor circuit ▪ Charge air pressure sensor failure	▪ Refer to the electrical circuit diagra and check the secondary compress outlet pressure sensor power and ground supplies for short circuit to ground, open circuit. Check the sig line for short circuit to ground, shor circuit to power, open circuit. Repa harness as required ▪ Check and install a new charge air pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0100-2A	Mass or Volume Air Flow Sensor "A" Circuit - Signal stuck in range	**NOTE:**
			Intake air temperature sensor is frozen within a plausible range
			▪ Intake air temperature sensor - combined with MAF sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Intake air temperature sensor - combined with MAF sensor failure
			  ▪ Contamination
			  ▪ Obstruction
				▪ Refer to the electrical circuit diagra and check the intake air temperatu sensor - combined with MAF senso circuit for short circuit to ground, sh circuit to power, open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new intake air temperature sensor - combined wit MAF sensor as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0102-16	Mass or Volume Air Flow Sensor "A" Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground
			▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Mass air flow and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Mass air flow sensor failure
			  ▪ Contamination
			  ▪ Obstruction
			▪ Mass air flow and temperature sensor failure
			  ▪ Contamination
			  ▪ Obstruction
				▪ Refer to the electrical circuit diagra and check the mass air flow sensor circuit for short circuit to ground, sh circuit to power, open circuit, high resistance
				▪ Refer to the electrical circuit diagra and check the mass air flow and temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new mass air flo sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'En control module mass air flow adaption clear' routine (0x0406
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatior routine (0x402C)
				▪ Check and install a new mass air flo and temperature sensor as requirec
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0103-17	Mass or Volume Air Flow Sensor "A" Circuit High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high	▪ Refer to the electrical circuit diagra and check the mass air flow sensor circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagra and check the mass air flow and temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c

			resistance
			▪ Mass air flow and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Mass air flow sensor failure
			  ▪ Contamination
			  ▪ Obstruction
			▪ Mass air flow and temperature sensor failure
			  ▪ Contamination
			  ▪ Obstruction
				corrosion
				▪ Check and install a new mass air flo sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'En control module mass air flow adaption clear' routine (0x0406
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatior routine (0x402C)
				▪ Check and install a new mass air flo and temperature sensor as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0105-84	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range
			▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Manifold absolute pressure and temperature sensor failure
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x0322 Corrected Intake Manif Absolute Pressure - KPa
				  ▪ 0x033E Boost Absolute Pressur Raw Value - KPa
				  ▪ 0x052C Manifold Absolute Pre Bank 1 - KPa
				  ▪ 0xF40B Intake Manifold Absolu Pressure - KPa
				▪ Refer to the electrical circuit diagra and check the manifold absolute pressure and temperature sensor ci for short circuit to ground
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new manifold absolute pressure and temperature sensor as required
				▪ Using the manufacturer approved

				diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0105-85	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range
			▪ Manifold absolute pressure and temperature sensor circuit short circuit to power
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Manifold absolute pressure and temperature sensor failure
				▪ Diagnosis of this DTC may require
				the manufacturer approved diagno
				system check datalogger signals
				  ▪ 0x0322 Corrected Intake Manif
				  Absolute Pressure - KPa
				  ▪ 0x033E Boost Absolute Pressu
				  Raw Value - KPa
				  ▪ 0x052C Manifold Absolute Pre
				  Bank 1 - KPa
				  ▪ 0xF40B Intake Manifold Absolu
				  Pressure - KPa
				▪ Refer to the electrical circuit diagra
				and check the manifold absolute
				pressure and temperature sensor c
				for short circuit to power
				▪ Inspect connectors for signs of wat
				ingress, and pins for damage and/c
				corrosion
				▪ Check and install a new manifold
				absolute pressure and temperature
				sensor as required
				▪ Using the manufacturer approved
				diagnostic system clear all stored D
				using the Diagnosis Menu tab and
				retest

	P0106-84	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit Range/Performance - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range
			▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground
			▪ Connector is disconnected, connector pin is backed out,
				▪ Diagnosis of this DTC may require
				the manufacturer approved diagno
				system check datalogger signals
				  ▪ 0x0322 Corrected Intake Manif
				  Absolute Pressure - KPa
				  ▪ 0x033E Boost Absolute Pressu
				  Raw Value - KPa
				  ▪ 0x052C Manifold Absolute Pre
				  Bank 1 - KPa
				  ▪ 0xF40B Intake Manifold Absolu
				  Pressure - KPa
				▪ Refer to the electrical circuit diagra
				and check the manifold absolute
				pressure and temperature sensor c
				for short circuit to ground

			connector pin corrosion ▪ Manifold absolute pressure and temperature sensor failure	▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new manifold absolute pressure and temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0106-85	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit Range/Performance - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range
			▪ Manifold absolute pressure and temperature sensor circuit short circuit to power
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Manifold absolute pressure and temperature sensor failure
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x0322 Corrected Intake Manif Absolute Pressure - KPa
				  ▪ 0x033E Boost Absolute Pressur Raw Value - KPa
				  ▪ 0x052C Manifold Absolute Pre Bank 1 - KPa
				  ▪ 0xF40B Intake Manifold Absolu Pressure - KPa
				▪ Refer to the electrical circuit diagra and check the manifold absolute pressure and temperature sensor ci for short circuit to power
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new manifold absolute pressure and temperature sensor as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0107-16	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground
			▪ Manifold absolute pressure and
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x0322 Corrected Intake Manif Absolute Pressure - KPa
				  ▪ 0x033E Boost Absolute Pressur Raw Value - KPa
				  ▪ 0x052C Manifold Absolute Pre Bank 1 - KPa

			▪ temperature sensor circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Manifold absolute pressure and temperature sensor failure	▪ 0xF40B Intake Manifold Absolu Pressure - KPa ▪ Refer to the electrical circuit diagra and check the manifold absolute pressure and temperature sensor ci for short circuit to ground ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new manifold absolute pressure and temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0108-17	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Manifold absolute pressure and temperature sensor circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Manifold absolute pressure and temperature sensor failure	▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x0322 Corrected Intake Manif Absolute Pressure - KPa ▪ 0x033E Boost Absolute Pressu Raw Value - KPa ▪ 0x052C Manifold Absolute Pre Bank 1 - KPa ▪ 0xF40B Intake Manifold Absolu Pressure - KPa ▪ Refer to the electrical circuit diagra and check the manifold absolute pressure and temperature sensor ci for short circuit to power ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new manifold absolute pressure and temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P010C-16	Mass or Volume Air Flow Sensor "B" Circuit Low - Circuit voltage below	▪ The engine control module measured a voltage below a	▪ Refer to the electrical circuit diagra and check the mass air flow sensor circuit for short circuit to ground, sh

		threshold	specified range but not necessarily a short circuit to ground
			▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Mass air flow sensor failure
			  ▪ Contamination
			  ▪ Obstruction
				circuit to power, open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new mass air flc sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'En control module mass air flow adaption clear' routine (0x0406
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatior routine (0x402C)
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P010D-17	Mass or Volume Air Flow Sensor "B" Circuit High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power
			▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Mass air flow sensor failure
			  ▪ Contamination
			  ▪ Obstruction
				▪ Refer to the electrical circuit diagra and check the mass air flow sensor circuit for short circuit to ground, sl circuit to power, open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new mass air flc sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'En control module mass air flow adaption clear' routine (0x0406
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatior routine (0x402C)
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0110-00	Intake Air Temperature Sensor 1 Circuit Bank 1 -		▪ Refer to the electrical circuit diagra and check the intake air temperatu

		No sub type information	**NOTE:**
			Intake air temperature sensor is frozen within a plausible range
			▪ Intake air temperature sensor - combined with MAF sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Intake air temperature sensor - combined with MAF sensor failure
			  ▪ Contamination
			  ▪ Obstruction
				sensor - combined with MAF senso circuit for short circuit to ground, sh circuit to power, open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new intake air temperature sensor - combined wit MAF sensor as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0110-62	Intake Air Temperature Sensor 1 Circuit Bank 1 - Signal compare failure	▪ Intake air temperature sensor - combined with MAF sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Charge air cooler temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Ambient air temperature sensor circuit short circuit to	▪ Refer to the electrical circuit diagra and check the intake air temperatu sensor - combined with MAF senso circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagra and check the charge air cooler temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance ▪ Refer to the electrical circuit diagra and check the ambient air tempera sensor circuit for short circuit to gro short circuit to power, open circuit, resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c

			ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Intake air temperature sensor - combined with MAF sensor failure
			  ▪ Contamination
			  ▪ Obstruction
			▪ Charge air cooler temperature sensor failure
			  ▪ Contamination
			  ▪ Obstruction
			▪ Ambient air temperature sensor failure
			  ▪ Contamination
			  ▪ Obstruction
				corrosion
				▪ Check and install a new intake air temperature sensor - combined wit MAF sensor as required
				▪ Check and install a new charge air cooler temperature sensor as requi
				▪ Check and install a new ambient ai temperature sensor as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0111-84	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1 - Signal below allowable range	▪ Intake air temperature sensor - combined with MAF sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Intake air temperature sensor - combined with MAF sensor failure
			  ▪ Contamination
			  ▪ Obstruction
				▪ Refer to the electrical circuit diagra and check the intake air temperatu sensor - combined with MAF senso circuit for short circuit to ground, sh circuit to power, open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new intake air temperature sensor - combined wit MAF sensor as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0111-85	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1 - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range
			▪ Intake air temperature sensor - combined with MAF sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Intake air temperature sensor - combined with MAF sensor failure
			  ▪ Contamination
			  ▪ Obstruction
				▪ Refer to the electrical circuit diagra and check the intake air temperatu sensor - combined with MAF senso circuit for short circuit to ground, sh circuit to power, open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new intake air temperature sensor - combined wit MAF sensor as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0112-16	Intake Air Temperature Sensor 1 Circuit Low Bank 1 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Mass air flow and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Intake air temperature sensor - combined with mass air flow sensor	▪ Refer to the electrical circuit diagra and check the mass air flow and temperature sensor circuit for short circuit to ground, short circuit to pc open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new mass air flc and temperature sensor as requirec ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			- Connector is disconnected, connector pin is backed out, connector pin corrosion - Mass air flow and temperature sensor failure	
	P0113-17	Intake Air Temperature Sensor 1 Circuit High Bank 1 - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Mass air flow and temperature sensor circuit short circuit to power - Intake air temperature sensor - combined with mass air flow sensor - Connector is disconnected, connector pin is backed out, connector pin corrosion - Mass air flow and temperature sensor failure	- Refer to the electrical circuit diagra and check the mass air flow and temperature sensor circuit for short circuit to power - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new mass air flc and temperature sensor as requirec - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0116-62	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Signal compare failure	- Cooling system mechanical and fluid integrity - Cooling fan system mechanical integrity - Thermostat system mechanical integrity - Engine coolant temperature sensor 1 circuit short circuit to ground, short	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x0357 Engine Coolant Temperature Sensor - Volts 0xF405 Engine Coolant Temperature - Deg C - Check cooling system for mechanic and fluid integrity - Check cooling fan system for mechanical integrity

			circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine coolant temperature sensor 1 contamination ▪ Engine coolant temperature sensor 1 failure	▪ Check thermostat system for mechanical integrity ▪ Refer to the electrical circuit diagra and check the engine coolant temperature sensor 1 circuit for sho circuit to ground, short circuit to po open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new engine coo temperature sensor 1 as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0116-64	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Signal plausibility failure	▪ Cooling system mechanical and fluid integrity
			▪ Cooling fan system mechanical integrity
			▪ Thermostat system mechanical integrity
			▪ Engine coolant temperature sensor 1 circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Engine coolant temperature sensor 1 contamination
			▪ Engine coolant temperature sensor 1 failure
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x0357 Engine Coolant Temperature Sensor - Volts
				  ▪ 0xF405 Engine Coolant Temperature - Deg C
				▪ Check cooling system for mechanic and fluid integrity
				▪ Check cooling fan system for mechanical integrity
				▪ Check thermostat system for mechanical integrity
				▪ Refer to the electrical circuit diagra and check the engine coolant temperature sensor 1 circuit for sho circuit to ground, short circuit to po open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new engine coo temperature sensor 1 as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0116-84	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range
			▪ Cooling system mechanical and fluid integrity
			▪ Cooling fan system mechanical integrity
			▪ Thermostat system mechanical integrity
			▪ Engine coolant temperature sensor 1 circuit short circuit to ground, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Engine coolant temperature sensor 1 failure
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x0357 Engine Coolant Temperature Sensor - Volts
				  ▪ 0xF405 Engine Coolant Temperature - Deg C
				▪ Check cooling system for mechanic and fluid integrity
				▪ Check cooling fan system for mechanical integrity
				▪ Check thermostat system for mechanical integrity
				▪ Refer to the electrical circuit diagra and check the engine coolant temperature sensor 1 circuit for sho circuit to ground, open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion
				▪ Check and install a new engine coo temperature sensor 1 as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0116-85	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range
			▪ Cooling system mechanical and fluid integrity
			▪ Cooling fan system mechanical integrity
			▪ Thermostat system mechanical integrity
			▪ Engine coolant
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x0357 Engine Coolant Temperature Sensor - Volts
				  ▪ 0xF405 Engine Coolant Temperature - Deg C
				▪ Check cooling system for mechanic and fluid integrity
				▪ Check cooling fan system for mechanical integrity
				▪ Check thermostat system for mechanical integrity
				▪ Refer to the electrical circuit diagra and check the engine coolant temperature sensor 1 circuit for sho

			- temperature sensor 1 circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Engine coolant temperature sensor 1 failure	circuit to power - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new engine coo temperature sensor 1 as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0117-16	Engine Coolant Temperature Sensor 1 Circuit Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Engine coolant temperature sensor 1 circuit short circuit to ground, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Engine coolant temperature sensor 1 failure	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x0357 Engine Coolant Temperature Sensor - Volts 0xF405 Engine Coolant Temperature - Deg C - Refer to the electrical circuit diagra and check the engine coolant temperature sensor 1 circuit for sho circuit to ground, open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new engine coo temperature sensor 1 as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0118-17	Engine Coolant Temperature Sensor 1 Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Engine coolant temperature sensor 1 circuit short circuit to power - Connector is	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x0357 Engine Coolant Temperature Sensor - Volts 0xF405 Engine Coolant Temperature - Deg C - Refer to the electrical circuit diagra and check the engine coolant temperature sensor 1 circuit for sho circuit to power

			- disconnected, connector pin is backed out, connector pin corrosion - Engine coolant temperature sensor 1 failure	- Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new engine coo temperature sensor 1 as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0128-00	Coolant Thermostat (Coolant Temperature Below Thermostat Regulating Temperature) - No sub type information	**NOTE:** The cylinder head temperature does not reach its running temperature due to thermostat leakage - Thermostat is stuck open or is partially stuck open - Electrically heated thermostat circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Electrically heated thermostat failure	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x0392 Cylinder Head Tempera Sensor Voltage - Volts 0x0488 Engine Coolant Temperature Sensor #2 Voltage Volts 0x0489 Engine Coolant Temperature #2 - Deg C - Check thermostat operation - should open at 82degC only vi radiator out temperature (DIDx0489) 0x051D Cylinder Head Temper - Deg C - Check thermostat for mechanical integrity - Refer to the electrical circuit diagra and check the electrically heated thermostat circuit for short circuit t ground, short circuit to power, ope circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new electrically heated thermostat as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P012F-62	Engine Coolant Temperature / Engine Oil Temperature Correlation -	Radiator outlet temperature sensor circuit short circuit to	Refer to the electrical circuit diagra and check the radiator outlet temperature sensor circuit for short

	Signal compare failure	ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Radiator outlet temperature sensor failure	circuit to ground, short circuit to po open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new radiator ou temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

P0131-11	O2 Sensor Circuit Low Voltage Bank 1 Sensor 1 - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected
		▪ Fuel injection related DTCs
		▪ Air induction system leakage
		▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Heated oxygen sensor failure
			▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index
			▪ Check air induction system for leak
			▪ Refer to the electrical circuit diagra and check the heated oxygen sens heater circuit for short circuit to gro short circuit to power, open circuit, resistance
			▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
			▪ Check and install a new heated oxy sensor as required
			  ▪ Using the manufacturer approv diagnostic system carry out 'Oxygen sensor' routine (0x403
			  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatior routine (0x402C)
			▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

P0132-12	O2 Sensor Circuit High Voltage Bank 1 Sensor 1 - Circuit short To battery	▪ The engine control module has detected a vehicle power measurement	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index

			for a period longer than expected or has detected a vehicle power measurement when another value was expected
			▪ Fuel injection related DTCs
			▪ Air induction system leakage
			▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Heated oxygen sensor failure
				▪ Check air induction system for leak
				▪ Refer to the electrical circuit diagra and check the heated oxygen sens heater circuit for short circuit to gr short circuit to power, open circuit, resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new heated oxy sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Oxygen sensor' routine (0x403
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatio routine (0x402C)
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0134-00	O2 Sensor Circuit No Activity Detected Bank 1 Sensor 1 - No sub type information	▪ Fuel injection related DTCs ▪ Air induction system leakage ▪ Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Pre catalytic	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check air induction system for leak ▪ Refer to the electrical circuit diagra and check the exhaust gas tempera sensor - Pre catalytic converter circ for short circuit to ground, short cir to power, open circuit, high resistar ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new exhaust ga temperature sensor - Pre catalytic converter as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			converter failure	

	P0135-29	O2 Sensor Heater Circuit Bank 1 Sensor 1 - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions
			▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Heated oxygen sensor failure
				▪ Refer to the electrical circuit diagra and check the heated oxygen sens heater circuit for short circuit to gro short circuit to power, open circuit, resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new heated oxy sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Oxygen sensor' routine (0x403
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatio routine (0x402C)
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0148-91	Fuel Delivery Error - Parametric	▪ The engine control module has detected that the component parameter e.g. capacitance or inductance is outside its expected range ▪ This DTC is set as part of the high pressure fuel pump protection strategy. The DTC has set because the high pressure fuel pump is reaching a calibrated number of engine re-starts ▪ First level of high press fuel pump re-starts has been reached	▪ Check high pressure fuel pump for mechanical damage and excessive ▪ Check and install a new high pressu fuel pump as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the 'Diagnosis menu' tab and retest

	P0148-93	Fuel Delivery Error - No operation	- The engine control module has detected that the component is not operating - This DTC is set as part of the high pressure fuel pump protection strategy. The DTC has set because the high pressure fuel pump is reaching a calibrated number of engine re-starts - Second level of high press fuel pump re-starts has been reached	- Check high pressure fuel pump for mechanical damage and excessive - Check and install a new high pressu fuel pump as required - Using the manufacturer approved diagnostic system clear all stored D using the 'Diagnosis menu' tab and retest

	P014C-00	O2 Sensor Slow Response - Rich to Lean Bank 1 Sensor 1 - No sub type information	- Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
			- Connector is disconnected, connector pin is backed out, connector pin corrosion
			- Heated oxygen sensor failure
				- Refer to the electrical circuit diagra and check the heated oxygen sens heater circuit for short circuit to gro short circuit to power, open circuit, resistance
				- Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				- Check and install a new heated oxy sensor as required
				  - Using the manufacturer approv diagnostic system carry out 'Oxygen sensor' routine (0x403
				  - Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatio routine (0x402C)
				- Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P014D-00	O2 Sensor Slow Response - Lean to Rich Bank 1 Sensor 1 - No sub type information	- Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high	- Refer to the electrical circuit diagra and check the heated oxygen sens heater circuit for short circuit to gro short circuit to power, open circuit, resistance - Inspect connectors for signs of wat

			▪ resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Heated oxygen sensor failure
				ingress, and pins for damage and/c corrosion
				▪ Check and install a new heated oxy sensor as required
				  ▪ Using the manufacturer approv diagnostic system carry out 'Oxygen sensor' routine (0x403
				  ▪ Using the manufacturer approv diagnostic system carry out 'Di fueling adaption re-initializatio routine (0x402C)
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P016D-26	Excessive Time To Enter Closed Loop Fuel Pressure Control - Signal rate of change below threshold	▪ The signal transitions more slowly than is reasonably allowed ▪ Other fuel related DTCs ▪ Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance ▪ Crankshaft position sensor incorrectly installed ▪ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ▪ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out,	▪ Check engine control module for fu related DTCs and refer to relevant index ▪ Refer to the electrical circuit diagra and check connections are secure a wiring integrity ▪ Check crankshaft position sensor fo correct installation ▪ Check crankshaft position sensor ai gap to target rotor is correct, remo any debris from sensor face, check damaged teeth on rotor ▪ Refer to the electrical circuit diagra and check the crankshaft position sensor circuit for short circuit to gro short circuit to power, open circuit, resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check fuel rail system for mechanic integrity ▪ Check engine control module for mechanical integrity ▪ Check and install a new crankshaft position sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			connector pin corrosion ▪ Fuel rail system mechanical integrity ▪ Engine control module mechanical integrity ▪ Crankshaft position sensor failure	
	P0171-00	System Too Lean Bank 1 - No sub type information	▪ Other fuel injector related DTCs ▪ Air induction system leakage ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check air induction system for leak ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0172-00	System Too Rich Bank 1 - No sub type information	▪ Other fuel injector related DTCs ▪ Air induction system leakage ▪ Fuel injector is mechanically degraded	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check air induction system for leak ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D

				using the Diagnosis Menu tab and retest
	P017A-64	Cylinder Head Temperature Sensor Circuit - Signal plausibility failure	- Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Cylinder head temperature sensor failure - Engine coolant level or flow failure	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x051D Cylinder Head Temper - Deg C - Refer to the electrical circuit diagra and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new cylinder he temperature sensor as required - Refer to the workshop manual and check the engine cooling system to ensure the coolant condition, level flow is correct - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P017B-24	Cylinder Head Temperature Sensor Circuit Range/Performance - Signal stuck high	- The engine control module measures a signal that remains high when transitions are expected - Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Cylinder head	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x051D Cylinder Head Temper - Deg C - Refer to the electrical circuit diagra and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new cylinder he temperature sensor as required - Refer to the workshop manual and check the engine cooling system to ensure the coolant condition, level flow is correct

			▪ temperature sensor failure ▪ Engine coolant level or flow failure	▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P017B-2A	Cylinder Head Temperature Sensor Circuit Range/Performance - Signal stuck in range	▪ Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cylinder head temperature sensor failure ▪ Engine coolant level or flow failure	▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x051D Cylinder Head Temper - Deg C ▪ Refer to the electrical circuit diagra and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new cylinder he temperature sensor as required ▪ Refer to the workshop manual and check the engine cooling system to ensure the coolant condition, level flow is correct ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P017B-84	Cylinder Head Temperature Sensor Circuit Range/Performance - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Cooling system mechanical and fluid integrity ▪ Cooling fan system mechanical integrity ▪ Thermostat system mechanical integrity ▪ Cylinder head temperature sensor	▪ Check cooling system for mechanic and fluid integrity ▪ Check cooling fan system for mechanical integrity ▪ Check thermostat system for mechanical integrity ▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x051D Cylinder Head Temper - Deg C ▪ Refer to the electrical circuit diagra and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance ▪ Inspect connectors for signs of wat

			▪ circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cylinder head temperature sensor failure ▪ Engine coolant level or flow failure	ingress, and pins for damage and/c corrosion ▪ Check and install a new cylinder he temperature sensor as required ▪ Refer to the workshop manual and check the engine cooling system tc ensure the coolant condition, level flow is correct ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P017B-85	Cylinder Head Temperature Sensor Circuit Range/Performance - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Cooling system mechanical and fluid integrity ▪ Cooling fan system mechanical integrity ▪ Thermostat system mechanical integrity ▪ Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cylinder head temperature sensor failure	▪ Check cooling system for mechanic and fluid integrity ▪ Check cooling fan system for mechanical integrity ▪ Check thermostat system for mechanical integrity ▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x051D Cylinder Head Temper - Deg C ▪ Refer to the electrical circuit diagra and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to pc open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new cylinder he temperature sensor as required ▪ Refer to the workshop manual and check the engine cooling system tc ensure the coolant condition, level flow is correct ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			▪ Engine coolant level or flow failure	
	P017C-16	Cylinder Head Temperature Sensor Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Cooling system mechanical and fluid integrity ▪ Cooling fan system mechanical integrity ▪ Thermostat system mechanical integrity ▪ Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cylinder head temperature sensor failure ▪ Engine coolant level or flow failure	▪ Check cooling system for mechanic and fluid integrity ▪ Check cooling fan system for mechanical integrity ▪ Check thermostat system for mechanical integrity ▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x051D Cylinder Head Temper - Deg C ▪ Refer to the electrical circuit diagra and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new cylinder he temperature sensor as required ▪ Refer to the workshop manual and check the engine cooling system to ensure the coolant condition, level flow is correct ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P017D-17	Cylinder Head Temperature Sensor Circuit High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Cooling system mechanical and fluid integrity ▪ Cooling fan system	▪ Check cooling system for mechanic and fluid integrity ▪ Check cooling fan system for mechanical integrity ▪ Check thermostat system for mechanical integrity ▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals ▪ 0x051D Cylinder Head Temper

				- mechanical integrity - Thermostat system mechanical integrity - Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Cylinder head temperature sensor failure - Engine coolant level or flow failure	- Deg C - Refer to the electrical circuit diagra and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to po open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new cylinder he temperature sensor as required - Refer to the workshop manual and check the engine cooling system tc ensure the coolant condition, level flow is correct - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0180-2A	Fuel Temperature Sensor "A" Circuit - Signal stuck in range	- Fuel pump - high pressure - fuel temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel pump - high pressure - fuel temperature sensor failure	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x033F Fuel Rail Temperature Sensor Voltage - Volts 0x0522 Fuel Temperature "A" Deg C - Refer to the electrical circuit diagra and check the fuel pump - high pressure - fuel temperature sensor circuit for short circuit to ground, sl circuit to power, open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new fuel pump high pressure - fuel temperature se as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0180-62	Fuel Temperature Sensor "A" Circuit - Signal compare failure	▪ The engine control module detected failure when comparing two or more input parameters for plausibility
			▪ Fuel pump - high pressure - fuel temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Fuel pump - high pressure - fuel temperature sensor failure
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x033F Fuel Rail Temperature Sensor Voltage - Volts
				  ▪ 0x0522 Fuel Temperature "A" Deg C
				▪ Refer to the electrical circuit diagra and check the fuel pump - high pressure - fuel temperature sensor circuit for short circuit to ground, sh circuit to power, open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new fuel pump high pressure - fuel temperature se as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0181-84	Fuel Temperature Sensor "A" Circuit Range/Performance - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range
			▪ Fuel pump - high pressure - fuel temperature sensor circuit short circuit to ground
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Fuel pump - high pressure - fuel temperature sensor failure
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x033F Fuel Rail Temperature Sensor Voltage - Volts
				  ▪ 0x0522 Fuel Temperature "A" Deg C
				▪ Refer to the electrical circuit diagra and check the fuel pump - high pressure - fuel temperature sensor circuit for short circuit to ground
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new fuel pump high pressure - fuel temperature se as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

P0181-85	Fuel Temperature Sensor "A" Circuit Range/Performance - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range
		▪ Fuel pump - high pressure - fuel temperature sensor circuit short circuit to power
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Fuel pump - high pressure - fuel temperature sensor failure
			▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
			  ▪ 0x033F Fuel Rail Temperature Sensor Voltage - Volts
			  ▪ 0x0522 Fuel Temperature "A" Deg C
			▪ Refer to the electrical circuit diagra and check the fuel pump - high pressure - fuel temperature sensor circuit for short circuit to power
			▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
			▪ Check and install a new fuel pump high pressure - fuel temperature se as required
			▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

P0182-16	Fuel Temperature Sensor "A" Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground
		▪ Fuel pump - high pressure - fuel temperature sensor circuit short circuit to ground
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Fuel pump - high pressure - fuel temperature sensor failure
			▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
			  ▪ 0x033F Fuel Rail Temperature Sensor Voltage - Volts
			  ▪ 0x0522 Fuel Temperature "A" Deg C
			▪ Refer to the electrical circuit diagra and check the fuel pump - high pressure - fuel temperature sensor circuit for short circuit to ground
			▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
			▪ Check and install a new fuel pump high pressure - fuel temperature se as required
			▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

P0183-17	Fuel Temperature Sensor "A" Circuit High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power
		▪ Fuel pump - high pressure - fuel temperature sensor circuit short circuit to power
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Fuel pump - high pressure - fuel temperature sensor failure
			▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
			  ▪ 0x033F Fuel Rail Temperature Sensor Voltage - Volts
			  ▪ 0x0522 Fuel Temperature "A" Deg C
			▪ Refer to the electrical circuit diagra and check the fuel pump - high pressure - fuel temperature sensor circuit for short circuit to power
			▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
			▪ Check and install a new fuel pump high pressure - fuel temperature se as required
			▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

P0191-00	Fuel Rail Pressure Sensor A Circuit Range/Performance Bank 1 - No sub type information	▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Fuel rail pressure sensor failure
			▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
			  ▪ 0x0324 Fuel Rail Pressure Sens Volts
			  ▪ 0x03DC Fuel Rail Pressure - De - KPa
			  ▪ 0xF423 Fuel Rail Pressure - KPa
			▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground, sh circuit to power, open circuit, high resistance
			▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
			▪ Check and install a new fuel rail pressure sensor as required
			▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

P0191-	Fuel Rail Pressure Sensor A	▪ The engine control	▪ Diagnosis of this DTC may require

	16	Circuit Range/Performance Bank 1 - Circuit voltage below threshold	module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel rail pressure sensor failure	the manufacturer approved diagno system check datalogger signals ▪ 0x0324 Fuel Rail Pressure Sens Volts ▪ 0x03DC Fuel Rail Pressure - De - KPa ▪ 0xF423 Fuel Rail Pressure - KPa ▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new fuel rail pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0191-17	Fuel Rail Pressure Sensor A Circuit Range/Performance Bank 1 - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel rail pressure sensor failure	▪ Diagnosis of this DTC may require u the manufacturer approved diagno system check datalogger signals ▪ 0x0324 Fuel Rail Pressure Sens Volts ▪ 0x03DC Fuel Rail Pressure - De - KPa ▪ 0xF423 Fuel Rail Pressure - KPa ▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new fuel rail pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0191-84	Fuel Rail Pressure Sensor Circuit Range/Performance Bank 1 - Signal below allowable range	▪ Other fuel pressure related DTCs
			▪ Fuel system mechanical integrity
			▪ Fuel rail pressure sensor circuit short circuit to ground
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Fuel rail pressure sensor failure
				▪ Diagnosis of this DTC may require t the manufacturer approved diagno system check datalogger signals
				  ▪ 0x0324 Fuel Rail Pressure Sens Volts
				  ▪ 0x03DC Fuel Rail Pressure - De - KPa
				  ▪ 0xF423 Fuel Rail Pressure - KPa
				▪ Check fuel system for mechanical integrity
				▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index
				▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new fuel rail pressure sensor as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0191-85	Fuel Rail Pressure Sensor Circuit Range/Performance Bank 1 - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range
			▪ Other fuel pressure related DTCs
			▪ Fuel system mechanical integrity
			▪ Fuel rail pressure sensor circuit short circuit to power
			▪ Connector is disconnected, connector pin is backed out, connector pin
				▪ Diagnosis of this DTC may require t the manufacturer approved diagno system check datalogger signals
				  ▪ 0x0324 Fuel Rail Pressure Sens Volts
				  ▪ 0x03DC Fuel Rail Pressure - De - KPa
				  ▪ 0xF423 Fuel Rail Pressure - KPa
				▪ Check fuel system for mechanical integrity
				▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index
				▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to power
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion

		corrosion ▪ Fuel rail pressure sensor failure	▪ Check and install a new fuel rail pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

P0192-16	Fuel Rail Pressure Sensor A Circuit Low Bank 1 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground
		▪ Other fuel pressure related DTCs
		▪ Fuel rail pressure sensor circuit short circuit to ground
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Fuel rail pressure sensor failure
			NOTE:
			Operational requirements needec to allow the monitor to be fully tested. Drive the vehicle over 3 drive cycles (engine on/off) across varied driving conditions includin high torque to confirm the repair
			▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
			  ▪ 0x0324 Fuel Rail Pressure Sens Volts
			  ▪ 0x03DC Fuel Rail Pressure - De - KPa
			  ▪ 0xF423 Fuel Rail Pressure - KPa
			▪ Refer to the electrical circuit diagra and check connections are secure a wiring integrity
			▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index
			▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground
			▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
			▪ Check and install a new fuel rail pressure sensor as required
			▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

P0193-	Fuel Rail Pressure Sensor A	▪ The engine control	▪ Diagnosis of this DTC may require

	17	Circuit High - Circuit voltage above threshold	module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Other fuel pressure related DTCs ▪ Fuel rail pressure sensor circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel rail pressure sensor failure	the manufacturer approved diagno system check datalogger signals ▪ 0x0324 Fuel Rail Pressure Sens Volts ▪ 0x03DC Fuel Rail Pressure - De - KPa ▪ 0xF423 Fuel Rail Pressure - KPa ▪ Refer to the electrical circuit diagra and check connections are secure a wiring integrity ▪ Check engine control module for fu pressure related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to power ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel rail pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0196-29	Engine Oil Temperature Sensor "A" Range/Performance - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Oil level and temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Oil level and temperature sensor	▪ Refer to electrical circuit diagrams a check the oil level and temperature sensor circuit for short circuit to gro short circuit to power, open circuit, resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new oil level an temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			failure	
	P0201-00	Cylinder 1 Injector "A" Circuit - No sub type information	- Fuel injector circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel injector failure	- Refer to the electrical circuit diagra and check the fuel injector circuit fo open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new fuel injecto required - Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0202-00	Cylinder 2 Injector "A" Circuit - No sub type information	- Fuel injector circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel injector failure	- Refer to the electrical circuit diagra and check the fuel injector circuit fo open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new fuel injecto required - Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0203-00	Cylinder 3 Injector "A" Circuit - No sub type information	- Fuel injector circuit open circuit, high resistance - Connector is disconnected, connector pin is	- Refer to the electrical circuit diagra and check the fuel injector circuit fo open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion

			backed out, connector pin corrosion ▪ Fuel injector failure	▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0204-00	Cylinder 4 Injector "A" Circuit - No sub type information	▪ Fuel injector circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0205-00	Cylinder 5 Injector "A" Circuit - No sub type information	▪ Fuel injector circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes

				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0206-00	Cylinder 6 Injector "A" Circuit - No sub type information	▪ Fuel injector circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P020A-00	Cylinder 1 Injection Timing - No sub type information	▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P020B-00	Cylinder 2 Injection Timing - No sub type information	▪ Other fuel injector related DTCs ▪ Fuel injector is	▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto

			mechanically degraded ▪ Fuel injector is carbonized	required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P020C-00	Cylinder 3 Injection Timing - No sub type information	▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P020D-00	Cylinder 4 Injection Timing - No sub type information	▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P020E-00	Cylinder 5 Injection Timing - No sub type information	▪ Other fuel injector related DTCs ▪ Fuel injector is	▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto

			mechanically degraded ▪ Fuel injector is carbonized	required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P020F-00	Cylinder 6 Injection Timing - No sub type information	▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0216-00	Injector/Injection Timing Control Circuit - No sub type information	▪ Fuel pressure control valve mechanical integrity ▪ Fuel pressure control valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin	▪ Check fuel pressure control valve fo mechanical integrity ▪ Refer to the electrical circuit diagra and check the fuel pressure control valve circuit for short circuit to grou short circuit to power, open circuit, resistance ▪ Refer to the electrical circuit diagra and check the fuel rail pressure sen circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new fuel pressu control valve as required ▪ Check and install a new fuel rail pressure sensor as required

			corrosion ▪ Fuel pressure control valve failure ▪ Fuel rail pressure sensor failure	▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0219-95	Engine Overspeed Condition - Incorrect assembly	▪ The engine control module has detected that the component has been incorrectly installed e.g. hydraulic pipes crossed over, circuits cross wired or polarity errors ▪ The DTC is set if the engine speed goes above a set rpm for greater than 1 second ▪ Subsequent engine damage may have occurred from the overspeed condition ▪ Engine oil level ▪ Air induction system excessive oil ▪ Abusive driving style ▪ Crankshaft position sensor signal intermittent ▪ Crankshaft position sensor incorrectly installed ▪ Crankshaft position sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Crankshaft position sensor target rotor damaged ▪ Incorrect gear was selected	▪ Check and correct engine oil level ▪ Check air induction system for excessive oil ▪ Check crankshaft position sensor is correctly installed ▪ Refer to electrical circuit diagrams a check the crankshaft position senso circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Check crankshaft position sensor ta rotor for damage ▪ Using the manufacturer approved diagnostic system check transmissio control module for DTCs and refer the relevant DTC index ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			- Other transmission related DTCs - Automatic transmission failure	
	P0234-00	Turbocharger/Supercharger "A" Overboost Condition - No sub type information	- Excessive charged air pressure has been measured - Other related DTCs - Fuel injection related DTCs - Exhaust system leakage - Excessive engine oil level - Excessive oil in the air induction system - Air induction system leakage - Charge air pressure sensor blocked - Turbocharger system / vanes mechanical integrity	- Check engine control module for related DTCs and refer to relevant index - Check engine control module for fu injection related DTCs and refer to relevant DTC index - Check exhaust system for mechani integrity - Check and correct the engine oil le - Check air induction system / turbocharger for failure relating to excessive charged air conditions - Check air induction system for leak - Check air induction system for excessive oil - Check charge air pressure sensor fo blockage - Check turbocharger system / vanes mechanical integrity - Using the manufacturer approved diagnostic system perform routine Inline diagnostic unit 2 diagnostic t Turbocharger - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0234-77	Turbocharger/Supercharger "A" Overboost Condition - Commanded position not reachable		- Check the engine control module f related DTCs and if DTCs **P2463-0 and P246B-00** are also logged, go Pinpoint test A For additional information, refer to **Diesel Particulate Filter - TDV6 3. Diesel - Gen 2** (309-00A Exhaust System - TDV6 3.0L Diesel, Diagno and Testing). - Check engine control module for fu injection related DTCs and refer to relevant DTC index

			NOTE: A customer may express a concern that the amber Diesel Particulate Filter (DPF) warning indicator is illuminated on the Instrument Cluster (IC) requesting a DPF regeneration. If the regeneration is not completed the DPF warning indicator will then turn red. The DTCs **P2463-00 and P246B-00** will be stored in the Engine Control Module (ECM) memory ▪ The electric throttle air flow characteristics are incompatible with the engine calibration ▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment	▪ Check exhaust system for mechani integrity ▪ Check and correct the engine oil le ▪ Check air induction system / turbocharger for failure relating to excessive charged air conditions ▪ Check air induction system for leak ▪ Check air induction system for excessive oil ▪ Check charge air pressure sensor fo blockage ▪ Check turbocharger system / vanes mechanical integrity ▪ Check electric throttle system / bla for mechanical integrity ▪ Using the manufacturer approved diagnostic system perform routine Inline diagnostic unit 2 diagnostic t Turbocharger ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			- Excessive charged air pressure has been measured - Other related DTCs - Fuel injection related DTCs - Exhaust system leakage - Excessive engine oil level - Excessive oil in the air induction system - Air induction system leakage - Charge air pressure sensor blocked - Turbocharger system / vanes mechanical integrity - Electric throttle system / blade mechanical integrity	
	P0235-94	Turbocharger/Supercharger Boost Sensor "A" Circuit - Unexpected operation	- The engine control module has detected that the component is operating in a way or at a time that it has not been commanded to operate - Charge air shut-off valve is stuck open - Turbine intake shut-off valve	- Check the vacuum system, Charge shut-off valve and Turbine intake sh off valve operation. **Refer to sectio 303-04B or 303-04D and perform pinpoint test A Vacuum Control System Tests** - Using the manufacturer approved diagnostic system perform routine Inline diagnostic unit 2 diagnostic t Turbocharger - Using the Jaguar Land Rover Appro Diagnostic Equipment, clear all sto DTCs and retest

	P023A-13	Charge Air Cooler Coolant Pump Control Circuit/Open - Circuit open	- The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  - 0x05D6 Auxiliary Engine Coola Pump - Commanded - %
				  - 0x05DA Coolant Pump Variabl Flow Actuator Position - Measu %

			- Charge air coolant pump circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Charge air coolant pump failure	- Refer to the electrical circuit diagra and check the charge air coolant pu circuit for open circuit, high resistar - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new charge air coolant pump as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P023B-11	Charge Air Cooler Coolant Pump Control Circuit Low - Circuit short to ground	- The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Charge air coolant pump circuit short circuit to ground - Connector is disconnected, connector pin is backed out, connector pin corrosion - Charge air coolant pump failure	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x05D6 Auxiliary Engine Coola Pump - Commanded - % 0x05DA Coolant Pump Variabl Flow Actuator Position - Measu % - Refer to the electrical circuit diagra and check the charge air coolant pu circuit for short circuit to ground - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new charge air coolant pump as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P023C-12	Charge Air Cooler Coolant Pump Control Circuit High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Charge air coolant pump circuit short circuit to power	- Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals 0x05D6 Auxiliary Engine Coola Pump - Commanded - % 0x05DA Coolant Pump Variabl Flow Actuator Position - Measu % - Refer to the electrical circuit diagra and check the charge air coolant pu circuit for short circuit to power - Inspect connectors for signs of wat

			- Connector is disconnected, connector pin is backed out, connector pin corrosion - Charge air coolant pump failure	ingress, and pins for damage and/c corrosion - Check and install a new charge air coolant pump as required - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0241-16	Turbocharger/Supercharger Boost Sensor "B" Circuit Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Charge air pressure sensor - Secondary circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Charge air pressure sensor - Secondary failure	- Refer to the electrical circuit diagra and check the charge air pressure sensor - Secondary circuit for short circuit to ground, short circuit to pc open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new charge air pressure sensor - Secondary as req - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0242-17	Turbocharger/Supercharger Boost Sensor "B" Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Charge air pressure sensor - Secondary circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is	- Refer to the electrical circuit diagra and check the charge air pressure sensor - Secondary circuit for short circuit to ground, short circuit to pc open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new charge air pressure sensor - Secondary as req - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

<table>
<tr><th></th><th></th><th></th><th></th><th></th></tr>
<tr><td></td><td></td><td></td><td>disconnected, connector pin is backed out, connector pin corrosion
Charge air pressure sensor - Secondary failure</td><td></td></tr>
<tr><td></td><td>P0251-13</td><td>Injection Pump Fuel Metering Control "A" (Cam/Rotor/Injector) - Circuit open</td><td>The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output
Fuel pump - high pressure circuit open circuit, high resistance
Fuel volume control valve
Connector is disconnected, connector pin is backed out, connector pin corrosion
Fuel pump - high pressure failure
Fuel volume control valve</td><td>Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
0x03C2 Fuel Volume Control V Duty Cycle - %
0x03EA Fuel Volume Control V Current - Measured - Amp
Refer to the electrical circuit diagra and check the fuel pump - high pressure sensor circuit for open circ high resistance
Inspect connectors for signs of wat ingress, and pins for damage and/ corrosion
Check and install a new fuel pump high pressure as required
Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest</td></tr>
<tr><td></td><td>P0252-4B</td><td>Injection Pump Fuel Metering Control "A" Range/Performance (Cam/Rotor/Injector) - Over temperature</td><td>The engine control module detected an internal temperature above the expected range
Other fuel related DTCs
Fuel system mechanical failure, misfuelling
Fuel pump - high pressure circuit short circuit to power</td><td>Check engine control module for fu related DTCs and refer to relevant index
Check fuel system for mechanical integrity
Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
0x03C2 Fuel Volume Control V Duty Cycle - %
0x03EA Fuel Volume Control V Current - Measured - Amp</td></tr>
</table>

			▪ Fuel volume control valve
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Fuel pump - high pressure failure
			  ▪ Fuel volume control valve
				▪ Refer to the electrical circuit diagra and check the fuel pump - high pressure sensor circuit for short circ to power
				  ▪ Fuel volume control valve
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new fuel pump high pressure as required
				  ▪ Fuel volume control valve
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0253-11	Injection Pump Fuel Metering Control "A" Low (Cam/Rotor/Injector) - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected
			▪ Fuel pump - high pressure circuit short circuit to ground
			  ▪ Fuel volume control valve
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Fuel pump - high pressure failure
			  ▪ Fuel volume control valve
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x03C2 Fuel Volume Control V Duty Cycle - %
				  ▪ 0x03EA Fuel Volume Control V Current - Measured - Amp
				▪ Refer to the electrical circuit diagra and check the fuel pump - high pressure sensor circuit for short circ to ground
				  ▪ Fuel volume control valve
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new fuel pump high pressure as required
				  ▪ Fuel volume control valve
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0254-12	Injection Pump Fuel Metering Control "A" High (Cam/Rotor/Injector) - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has	▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x03C2 Fuel Volume Control V Duty Cycle - %
				  ▪ 0x03EA Fuel Volume Control V

			detected a vehicle power measurement when another value was expected
			▪ Fuel pump - high pressure circuit short circuit to power
			  ▪ Fuel volume control valve
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Fuel pump - high pressure failure
			  ▪ Fuel volume control valve
				Current - Measured - Amp
				▪ Refer to the electrical circuit diagra and check the fuel pump - high pressure sensor circuit for short circ to power
				  ▪ Fuel volume control valve
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new fuel pump high pressure as required
				  ▪ Fuel volume control valve
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0255-1F	Injection Pump Fuel Metering Control "A" Intermittent (Cam/Rotor/Injector) - Circuit intermittent	▪ Fuel pump - high pressure circuit short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Fuel volume control valve
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Fuel pump - high pressure failure
			  ▪ Fuel volume control valve
				▪ Diagnosis of this DTC may require the manufacturer approved diagno system check datalogger signals
				  ▪ 0x03C2 Fuel Volume Control V Duty Cycle - %
				  ▪ 0x03EA Fuel Volume Control V Current - Measured - Amp
				▪ Refer to the electrical circuit diagra and check the fuel pump - high pressure sensor circuit for short circ to ground, short circuit to power, o circuit, high resistance
				  ▪ Fuel volume control valve
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new fuel pump high pressure as required
				  ▪ Fuel volume control valve
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0261-00	Cylinder 1 Injector "A" Circuit Low - No sub type information	▪ Low side current of affected cylinder is below a threshold	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, short circuit

			▪ while high side current of another cylinder on this bank is above a second threshold ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0261-11	Cylinder 1 Injector "A" Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Driver output voltage of fuel injector below threshold ▪ Fuel injector circuit short circuit to ground, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0261-23	Cylinder 1 Injector Circuit "A" Low - Signal stuck low	▪ The engine control module measures a signal that remains	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista

			▪ low when transitions are expected ▪ Signal value below threshold ▪ Fuel injector circuit short circuit to ground, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0262-00	Cylinder 1 Injector "A" Circuit High - No sub type information	▪ Low side current of affected cylinder is below a threshold while high side current of another cylinder on this bank is above a second threshold ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, short circuit power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0263-92	Cylinder 1 Contribution/Balance - Performance or incorrect operation	▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ Other fuel injector	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out

			related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0264-00	Cylinder 2 Injector "A" Circuit Low - No sub type information	▪ Harness failure - Fuel injector control circuit short circuit ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the injector control circu between the engine control modul and the cylinder 2 fuel injector for s circuit between the two wires ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0264-11	Cylinder 2 Injector "A" Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Driver output voltage of fuel injector below threshold ▪ Fuel injector circuit short circuit to ground, high resistance ▪ Connector is disconnected,	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			▪ connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	
	P0264-23	Cylinder 2 Injector "A" Circuit Low - Signal stuck low	▪ The engine control module measures a signal that remains low when transitions are expected ▪ Signal value below threshold ▪ Fuel injector circuit short circuit to ground, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0265-00	Cylinder 2 Injector "A" Circuit High - No sub type information	▪ Low side current of affected cylinder is below a threshold while high side current of another cylinder on this bank is above a second threshold ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, short circuit power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0266-92	Cylinder 2 Contribution/Balance - Performance or incorrect operation	- The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - Other fuel injector related DTCs - Fuel injector is mechanically degraded - Fuel injector is carbonized	- Check engine control module for fu injection related DTCs and refer to relevant DTC index - Check fuel injector for mechanical integrity - Check and install a new fuel injecto required - Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0267-00	Cylinder 3 Injector "A" Circuit Low - No sub type information	- Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel injector failure	- Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, short circuit power, open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion - Check and install a new fuel injecto required - Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0267-11	Cylinder 3 Injector "A" Circuit Low - Circuit short to ground	- The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was	- Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista - Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion - Check and install a new fuel injecto required

			expected ▪ Driver output voltage of fuel injector below threshold ▪ Fuel injector circuit short circuit to ground, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0267-23	Cylinder 3 Injector "A" Circuit Low - Signal stuck low	▪ The engine control module measures a signal that remains low when transitions are expected ▪ Signal value below threshold ▪ Fuel injector circuit short circuit to ground, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0268-00	Cylinder 3 Injector "A" Circuit High - No sub type information	▪ Low side current of affected cylinder is below a threshold while high side current of another cylinder on this bank is above a second threshold ▪ Fuel injector circuit short circuit to ground, short circuit	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, short circuit power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv

			to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0269-00	Cylinder 3 Contribution/Balance - No sub type information	▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P026A-77	Charge Air Cooler Efficiency Below Threshold - Commanded position not reachable	▪ Charge air cooler system mechanical integrity ▪ Cooling system mechanical and fluid integrity ▪ Charge air coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Check charge air cooler system for mechanical integrity ▪ Check cooling system for mechanic and fluid integrity ▪ Refer to the electrical circuit diagra and check the charge air coolant se circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new charge air coolant pump as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and

			▪ Charge air coolant pump failure	retest
	P026B-00	Injection Timing Performance - No sub type information	▪ Fuel injector powerstage overheated (internal to engine control module) ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module and surrounding area for excessive / abnormal heat source ▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P026C-84	Fuel Injection Quantity Lower Than Expected - Signal below allowable range	▪ Other fuel injector related DTCs ▪ Air induction system leakage ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check air induction system for leak ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P026D-85	Fuel Injection Quantity Higher Than Expected -	▪ The engine control module has	▪ Check engine control module for fu injection related DTCs and refer to

		Signal above allowable range	determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Other fuel injector related DTCs ▪ Air induction system leakage ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	relevant DTC index ▪ Check air induction system for leak ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P026E-00	Charge Air Cooler Coolant Pump Performance - No sub type information	▪ Charge air cooler system mechanical integrity ▪ Cooling system mechanical and fluid integrity ▪ Charge air coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Charge air coolant pump failure	▪ Check charge air cooler system for mechanical integrity ▪ Check cooling system for mechanic and fluid integrity ▪ Refer to the electrical circuit diagra and check the charge air coolant se circuit for short circuit to ground, sh circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new charge air coolant pump as required ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P026E-7B	Charge Air Cooler Coolant Pump Performance - Low fluid level	▪ The engine control module has detected that a fluid level is too low for proper operation of the system ▪ Charge air cooler system mechanical	▪ Check charge air cooler system for mechanical integrity ▪ Check cooling system for mechanic and fluid integrity ▪ Refer to the electrical circuit diagra and check the charge air coolant pu circuit for short circuit to ground, sh circuit to power, open circuit, high

			integrity
			▪ Cooling system mechanical and fluid integrity
			  ▪ Air in system
			  ▪ Fluid level low
			▪ Charge air coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Charge air coolant pump failure
				resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new charge air coolant pump as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P026E-92	Charge Air Cooler Coolant Pump Performance - Performance or incorrect operation	▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way
			▪ Charge air cooler system mechanical integrity
			▪ Cooling system mechanical and fluid integrity
			  ▪ Air in system
			  ▪ Fluid level low
			▪ Charge air coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Dry running pump
			  ▪ Frozen coolant
			  ▪ Debris in coolant
				▪ Check charge air cooler system for mechanical integrity
				▪ Check cooling system for mechanic and fluid integrity
				▪ Refer to the electrical circuit diagra and check the charge air coolant pu circuit for short circuit to ground, sh circuit to power, open circuit, high resistance
				▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				▪ Check and install a new charge air coolant pump as required
				▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			- Seized bearings
			- Connector is disconnected, connector pin is backed out, connector pin corrosion
			- Charge air coolant pump failure
			  - Dry running pump
			  - Frozen coolant
			  - Debris in coolant
			  - Seized bearings	

	P026E-98	Charge Air Cooler Coolant Pump Performance - Component or system over temperature	- The engine control module has detected that the temperature is too high for the correct operation of the component or system
			- Charge air cooler system mechanical integrity
			- Cooling system mechanical and fluid integrity
			  - Air in system
			  - Fluid level low
			- Charge air coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance
			  - Dry running pump
			  - Frozen coolant
			  - Debris in coolant
			  - Seized bearings
			- Connector is disconnected, connector pin is backed out,
				- Check charge air cooler system for mechanical integrity
				- Check cooling system for mechanic and fluid integrity
				- Refer to the electrical circuit diagra and check the charge air coolant pu circuit for short circuit to ground, sh circuit to power, open circuit, high resistance
				- Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion
				- Check and install a new charge air coolant pump as required
				- Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

			connector pin corrosion
			▪ Charge air coolant pump failure
			  ▪ Dry running pump
			  ▪ Frozen coolant
			  ▪ Debris in coolant
			  ▪ Seized bearings	

	P0270-00	Cylinder 4 Injector "A" Circuit Low - No sub type information	▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, short circuit power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0270-11	Cylinder 4 Injector "A" Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Driver output voltage of fuel injector below threshold ▪ Fuel injector circuit short circuit to	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved

			- ground, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel injector failure	diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0270-23	Cylinder 4 Injector "A" Circuit Low - Signal stuck low	- The engine control module measures a signal that remains low when transitions are expected - Signal value below threshold - Fuel injector circuit short circuit to ground, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel injector failure	- Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new fuel injecto required - Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0271-00	Cylinder 4 Injector "A" Circuit High - No sub type information	- Low side current of affected cylinder is below a threshold while high side current of another cylinder on this bank is above a second threshold - Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out,	- Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, short circuit power, open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion - Check and install a new fuel injecto required - Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored D

			connector pin corrosion ▪ Fuel injector failure	using the Diagnosis Menu tab and retest
	P0272-92	Cylinder 4 Contribution/Balance - Performance or incorrect operation	▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0273-00	Cylinder 5 Injector "A" Circuit Low - No sub type information	▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, short circuit power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0273-11	Cylinder 5 Injector "A" Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista

			period longer than expected or has detected a ground measurement when another value was expected ▪ Driver output voltage of fuel injector below threshold ▪ Fuel injector circuit short circuit to ground, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0273-23	Cylinder 5 Injector "A" Circuit Low - Signal stuck low	▪ The engine control module measures a signal that remains low when transitions are expected ▪ Signal value below threshold ▪ Fuel injector circuit short circuit to ground, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fc short circuit to ground, high resista ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0274-00	Cylinder 5 Injector "A" Circuit High - No sub type information	▪ Low side current of affected cylinder is below a threshold while high side current of another cylinder on this bank	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fc short circuit to ground, short circuit power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c

			is above a second threshold - Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel injector failure	corrosion - Check and install a new fuel injecto required - Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0275-92	Cylinder 5 Contribution/Balance - Performance or incorrect operation	- The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - Other fuel injector related DTCs - Fuel injector is mechanically degraded - Fuel injector is carbonized	- Check engine control module for fu injection related DTCs and refer to relevant DTC index - Check fuel injector for mechanical integrity - Check and install a new fuel injecto required - Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0276-00	Cylinder 6 Injector "A" Circuit Low - No sub type information	- Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel injector failure	- Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, short circuit power, open circuit, high resistance - Inspect connectors for signs of wat ingress, and pins for damage and/o corrosion - Check and install a new fuel injecto required - Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine

				▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0276-11	Cylinder 6 Injector "A" Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Driver output voltage of fuel injector below threshold ▪ Fuel injector circuit short circuit to ground, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0276-23	Cylinder 6 Injector "A" Circuit Low - Signal stuck low	▪ The engine control module measures a signal that remains low when transitions are expected ▪ Signal value below threshold ▪ Fuel injector circuit short circuit to ground, high resistance ▪ Connector is disconnected,	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, high resista ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the

			connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0277-00	Cylinder 6 Injector "A" Circuit High - No sub type information	▪ Low side current of affected cylinder is below a threshold while high side current of another cylinder on this bank is above a second threshold ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Refer to the electrical circuit diagra and check the fuel injector circuit fo short circuit to ground, short circuit power, open circuit, high resistance ▪ Inspect connectors for signs of wat ingress, and pins for damage and/c corrosion ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P0278-00	Cylinder 6 Contribution/Balance - No sub type information	▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P0299-77	Turbocharger/Supercharger "A" Underboost Condition - Commanded position not reachable	▪ The boosting system is unable to build/control the boost pressure in the air induction system ▪ Other related DTCs ▪ Fuel injection related DTCs ▪ Exhaust system leakage ▪ Pre turbocharger ▪ Excessive oil in the air induction system ▪ Air induction system leakage ▪ Charge air pressure sensor blocked ▪ Charge air shut-off valve ▪ Turbine intake shut-off valve ▪ Turbocharger system / vanes mechanical integrity	▪ Check engine control module for related DTCs and refer to relevant index ▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check exhaust system for mechani integrity ▪ Check air induction system / turbocharger for failure relating to reduced charged air conditions ▪ Check air induction system for leak ▪ Check air induction system for excessive oil ▪ Check charge air pressure sensor fo blockage ▪ Check the vacuum system, Charge shut-off valve and Turbine intake sh off valve operation. **Refer to sectio 303-04B or 303-04D and perform pinpoint test A Vacuum Control System Tests** ▪ Check turbocharger system / vanes mechanical integrity ▪ Using the Jaguar Land Rover appro diagnostic equipment perform rout Inline diagnostic unit 2 diagnostic t Turbocharger ▪ Using the Jaguar Land Rover Appro Diagnostic Equipment, check and record value of datalogger signal, Diesel Particulate Filter Soot Concentration (0x042C). Please inc this value in the Analytical Warranty System technician verbatim ▪ Using the Jaguar Land Rover Appro Diagnostic Equipment, clear all sto DTCs and retest
	P02CC-35	Cylinder 1 Fuel Injector Offset Learning at Min Limit - Signal high time > maximum	▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Other fuel injector	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required

			▪ related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P02CD-32	Cylinder 1 Fuel Injector Offset Learning at Max Limit - Signal low time < minimum	▪ The engine control module detected the low pulse is too narrow with respect to time ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P02CE-35	Cylinder 2 Fuel Injector Offset Learning at Min Limit - Signal high time > maximum	▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

P02CF-32	Cylinder 2 Fuel Injector Offset Learning at Max Limit - Signal low time < minimum	▪ The engine control module detected the low pulse is too narrow with respect to time ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
P02D0-35	Cylinder 3 Fuel Injector Offset Learning at Min Limit - Signal high time > maximum	▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
P02D1-32	Cylinder 3 Fuel Injector Offset Learning at Max Limit - Signal low time < minimum	▪ The engine control module detected the low pulse is too narrow with respect to time ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out

			degraded ▪ Fuel injector is carbonized	'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P02D2-35	Cylinder 4 Fuel Injector Offset Learning at Min Limit - Signal high time > maximum	▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P02D3-32	Cylinder 4 Fuel Injector Offset Learning at Max Limit - Signal low time < minimum	▪ The engine control module detected the low pulse is too narrow with respect to time ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest

	P02D4-35	Cylinder 5 Fuel Injector Offset Learning at Min Limit - Signal high time > maximum	▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P02D5-32	Cylinder 5 Fuel Injector Offset Learning at Max Limit - Signal low time < minimum	▪ The engine control module detected the low pulse is too narrow with respect to time ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
	P02D6-35	Cylinder 6 Fuel Injector Offset Learning at Min Limit - Signal high time > maximum	▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically	▪ Check engine control module for fu injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injecto required ▪ Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine

		- degraded - Fuel injector is carbonized	- Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest
P02D7-32	Cylinder 6 Fuel Injector Offset Learning at Max Limit - Signal low time < minimum	- The engine control module detected the low pulse is too narrow with respect to time - Other fuel injector related DTCs - Fuel injector is mechanically degraded - Fuel injector is carbonized	- Check engine control module for fu injection related DTCs and refer to relevant DTC index - Check fuel injector for mechanical integrity - Check and install a new fuel injecto required - Using the manufacturer approv diagnostic system carry out 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored D using the Diagnosis Menu tab and retest